Optimizing Brain Tumor Segmentation by Multi-Slicing images with T1, T1CE, FLAIR Modalities using U-Net

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Abstract

To start, we're looking into brain tumors from the BraTS dataset, which has a large amount of MRI images stored free to use. MRI's are good at scanning and showing soft tissue, in this case the brain and its tumors. Instead of mapping out tumors from MRI scans manually, we can use deep learning in order to make it more quick and efficient. Our goal in this project is to determine the best modality for whole tumor segmentation using T1, T1CE, and FLAIR multi-slice images. We hope our results revealed which modality produced the best segmentation accuracy for brain tumors.

Background

- **U-Net (Fig. 2)** is a deep learning network commonly used for image segmentation tasks that includes an encoder to capture image features, a decoder to generate the segmentation mask, and skip connections to allow the decoder to use information from the encoder at different scales.
- Magnetic Resonance Imaging (MRI) images (**Fig. 1**):
  - **T1 images**: A type of MRI that provides information on the brain's anatomy.
  - **T1CE images**: Similar to T1, but with a contrast agent that highlights certain areas of the brain.
  - **FLAIR images**: A type of MRI that suppresses the signal from cerebrospinal fluid and highlights pathology in the brain.
- **Multi-slice images** are slices of brain image, with selecting the center slice and skipping several slices before and after to acquire the next in the sequence.

- **Jaccard/Dice coefficient**: Jaccard measures the similarity between predicted and ground truth segmentation masks as the ratio of their intersection to their union, with a range of 0 to 1, while Dice uses twice the ratio of the intersection to the sum of their areas.

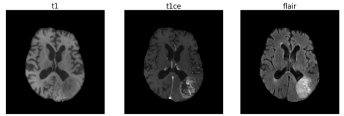

Fig. 1

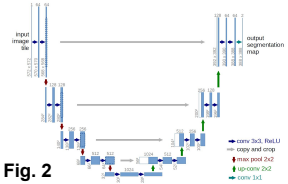

Fig. 2

Methods

- **Data**: BraTS dataset, which contains multi-modal MRI images of brain tumors.
- **Model**: U-Net, convolutional neural network that was developed for biomedical image segmentation.
- **Image Preprocessing**: We used multiple MRI image modalities including T1, T1CE, FLAIR while multi-slicing the images.
- **Training**: We trained the U-Net model using the T1, T1CE, FLAIR. We experimented the model by tuning hyper-parameters to find the best performing model.
- **Evaluation**: We evaluated the performance of the model using Jaccard and Dice coefficient metrics to measure the similarity between predicted and ground truth segmentation masks.

Results

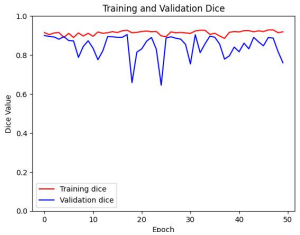

Fig. 3 FLAIR Dice

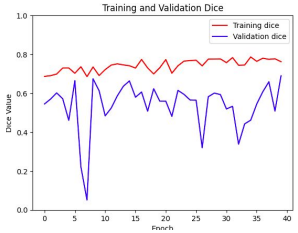

Fig. 4 T1 Dice

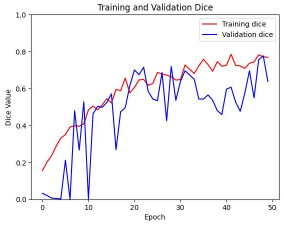

Fig. 5 T1CE Dice

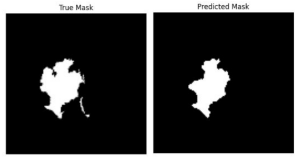

Fig. 6 FLAIR

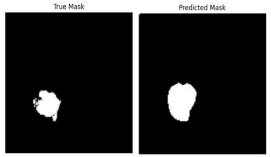

Fig. 7 T1

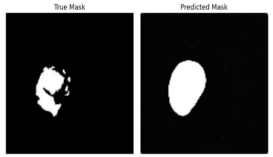

Fig. 8 T1CE

Conclusion

- In the end, the model with the best performance is **FLAIR (Fig. 3, Fig 6))** with the loss dice using these hyperparameters: **Learning rate**=0.00005, **Batch size** = 4, **Buffer size** = 1000, **Epoch** = 50 ,**Early stopping** = No, **Images acquired before/after middle slice** = 2 (5 total per patient id), **Image spacing** = 4
- We hope for further study to do multi-segmentation which can highlight and segment out each individual part of the tumor, such as necrosis, edema, & enhancing tumor parts from the whole.

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